Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/29/2019
PI Name:	Keller, Nancy Ph.D.		
Project Title:	“Omics” Data Mining of the ISS Aspergillus fumigatus Strains in Elucidating Virulence Characteristics		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53706-1313	**Congressional District:**	2
Comments:			
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System
Start Date:	02/01/2017	**End Date:**	01/31/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: Extended to 1/31/2019 per NSSC information; original end date was 1/31/2018 (Ed., 1/16/18)		
Key Personnel Changes/Previous PI:	January 2018 report: Dr. Ben Knox from Keller lab graduated in August 2017. He worked until Dec 2017 as a post doc on this project and then obtained a permanent job in Dec 2017. Graduate student Nicholas Raffa took his place.		
COI Name (Institution):	Venkateswaran , Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	NNX17AE50G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed Aspergillus fumigatus data mining (AFDM) project is a ground-based hypothesis-driven research investigation, derived from analyzing the “omics” data deposited in the GeneLab Data System (GLDS) that would translate spaceflight derived data into new knowledge. As A. fumigatus is an opportunistic pathogen causing pathologies ranging from allergic asthma to invasive aspergillosis, we assessed several characteristics of the International Space Station (ISS) isolates in comparison to two experimentally established clinical isolates, Af293 and CEA10. Virulence assessment in a larval zebrafish model of invasive aspergillosis revealed both ISSFT-021 and IF1SW-F4 as significantly more lethal compared to both Af293 and CEA10. Additionally, both ISSFT-021 and IF1SW-F4 exhibited significantly greater resistance to UV254 doses when compared to clinical isolates. We propose to assess that the enhanced virulence of the two ISS strains is mediated in part by up-regulation of spore pigments as a protective response to enhanced radiation in space. In addition to the “omics” data of the A. fumigatus strains deposited in the GLDS, the AFDM project will combine and compare the ground-based omics data that are either publicly available or generated by the Principal Investigator’s lab. The AFDM project will contribute to the ROSbio NASA Research Announcement NNH16ZTT001N-GL research emphasis (Systems Biology Experimental Research Using the GeneLab Data System) by generating new “omics” data about the fungal biology and adding it to the GeneLab for the benefit of future open science-based investigations in Earth. This will enable development of rapid microbial monitoring systems, countermeasures, novel pharmaceutical compounds, vaccine production, etc. The translational research aspects of the AFDM project are focusing the high-priority recommendations of the National Research Council's (NRC) Decadal Survey (P1. Establish a microbial observatory program on the ISS to conduct long-term, multigenerational studies of microbial population dynamics). The AFDM project will address specific NASA space biology-related questions by (a) investigating spaceflight omics data with specific aims to benefit human health (Aims 1 and 2), (b) addressing effects of spaceflight conditions on dynamics of microbes (fungi) with respect to cell processes such as understanding physiological responses (secondary metabolite production, Aims 2 and 3), UV resistance (Aim 3), and to the multiple stimuli encountered in spaceflight environments, and (c) characterizing the molecular responses in space grown fungi to pathogenesis (Aim 4). The objective is to provide NASA’s Space Biology program with “omics” information on how alterations in gravity affect genetic, mutational, metabolomic, and virulence characteristics in fungal species. By datamining the “omics” of two ISS A. fumigatus strains, we will address the NASA’s Space Biology Program science element (microbiology) and its guiding questions: (a) How the genetic, molecular, and biochemical processes of the ISS A. fumigatus strains are influenced by the space environment and (b) What systems biology mechanisms and pathways are responsible for the enhanced virulence in ISS isolates (NASA Task report 2016; Bacterial, Archaeal, & Fungal Diversity of the ISS--HEPA Filter System; 19-12829-27 under Task Order NNN13D111T; Principal Investigator Kasthuri Venkateswaran) when compared to the well characterized clinical isolates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In the Aspergillus fumigatus datamining (AFDM) project, we are mining the “omics” data generated and deposited in the NASA GeneLab. In addition, we are performing biological research intended at advancing future human exploration missions. The AFDM project is related to fundamental research–gaining knowledge of spaceflight alterations in the fungi isolated from ISS to improve life on Earth. Understanding the molecular mechanisms in the spaceflight fungi might reveal the presence of potential stress-induced biomolecules that are essential to adapt to microgravity conditions. Such stress-induced biological system modifications could be identified and applied (early diagnostics and superior countermeasure development) to improve human health in immunocompromised subjects on Earth (children, the aged, and people living in closed systems, etc.), as well as recognize secondary metabolites that are useful compounds for the biotech industry (antibacterial, novel pharmaceuticals, etc.). Similarly, identification of stress-induced biomolecules that are antimicrobials will facilitate maintaining crew health and their closed habitat system for the future human exploration. The AFDM project is focusing on: a. research aimed toward enabling the future of human exploration and b. on specific Earth-based disease (aspergillosis) pathways to gain insight from spaceflight “omics” data that may lead to new knowledge, treatments, and rapid diagnostic methods that will also improve life on Earth. The outcome of the AFDM project will serve “dual-use” (Earth and space) purpose by investigating the common underlying molecular networks, pathways, and mechanisms of life in space-exposed fungi (and compare with Earth counterparts), which are important to understanding human health and environment in space, and will translate space-derived knowledge to address specific human health conditions and environments here on Earth. Therefore, spaceflight fungal “omics” data and ground-based investigations with direct translational research connections of the AFDM project will directly address the important priority of the NASA Space Biology mission.
Task Progress:	Summary. Two Aspergillus fumigatus strains, ISSFT-021 and IF1SW-F4, were isolated from a HEPA filter and cupola wall of the International Space Station (ISS). As A. fumigatusis is an opportunistic pathogen causing pathologies ranging from allergic asthma to invasive aspergillosis, we assessed several characteristics of the ISS isolates in comparison to two experimentally established clinical isolates, Af293 and CEA10. Virulence assessment in a larval zebrafish model of invasive aspergillosis revealed both ISSFT-021 and IF1SW-F4 as significantly more lethal compared to both Af293 and CEA10 (Knox BP, Blachowicz A, Palmer JM, Romsdahl J, Huttenlocher A, Wang CC, Keller NP, Venkateswaran K. Aspergillus fumigatusisolates from air and surfaces of the International Space Station. mSphere. 2016 Oct 26;1(5):e00227-16). Additionally, both ISSFT-021 and IF1SW-F4 exhibited significantly greater resistance to UV254 doses when compared to clinical isolates (Blachowicz A, Knox BP, Romsdah lJ, Palmer JM, Huttenlocher A, Wang CCC, Keller NP, Venkateswaran K. 2016. Characterization of Aspergillus fumigatus isolated from air and surfaces of the International Space Station, 13th European Conference on Fungal Genetics, Paris, France, April 3-6, 2016). We hypothesized that the enhanced virulence of the two ISS strains is mediated in part by up-regulation of spore pigments as a protective response to enhanced radiation in space. Four spore pigments are known to contribute to virulence or toxicity (e.g., DH melanin, endocrocin, trypacidin, and fumigaclavine) and DH melanin is critical for UV protection for the fungus (the other pigments have not been assessed). The following two approaches were taken to address this hypothesis: Spore pigments contribute to UV protection and virulence depending on fungal strain. We created an isogenic set of 6 strains differing only in presence of the synthase gene required to produce one spore metabolite and determined UV radiation sensitivity. Enhanced UV resistance in evolution test. A controlled evolution study during which conidia of Af293 are exposed to sub-lethal doses of UV-C for 25 generations was conducted. We presented a research poster titled “Investigation of enhanced virulence mechanisms in Aspergillus fumigatus

	isolates from the International Space Station" at the 30th Fungal Genetics Conference held in Pacific Grove, CA from March 12-17, 2019. Blachowicz A, Raffa N, Knox BP, Huttenlocher A, Keller NP, Venkateswaran K, Wang CCC. 2019. Investigation of enhanced virulence mechanisms in Aspergillus fumigatus isolates from the International Space Station. Fungal Genetics Conference, Asilomar, Ca. [Ed. note: compiled with PI approval from PI's final report submitted March 2019]
Bibliography Type:	Description: (Last Updated: 06/25/2025)
Articles in Peer-reviewed Journals	Blachowicz A, Raffa N, Bok JW, Choera T, Knox B, Lim FY, Huttenlocher A, Wang CCC, Venkateswaran K, Keller NP. "Contributions of spore secondary metabolites to UV-C protection and virulence vary in different Aspergillus fumigatus strains." MBio. 2020 Feb 18;11(1):e03415-19. https://dx.doi.org/10.1128/mBio.03415-19 ; PubMed PMID: 32071276 ; PubMed Central PMCID: PMC7029147 , Feb-2020
Articles in Peer-reviewed Journals	Rosowski EE, Raffa N, Knox BP, Golenberg N, Keller NP, Huttenlocher A. "Macrophages inhibit Aspergillus fumigatus germination and neutrophil-mediated fungal killing." PLoS Pathog. 2018 Aug 2;14(8):e1007229. https://doi.org/10.1371/journal.ppat.1007229 ; PMID: 30071103; PMCID: PMC6091969 , Aug-2018